B.Tech CSE - Third Year | Lab 7

Assignment 7: Practice Set • *Aim: Practice array searching, sorting, multiplication and conditionals.*

Experiment No. 7.1: Search for 52 using a loop

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_01_Search_for_52_using_a_loop {
    public static void main(String[] args) {
        int[] a = {45, 52, 80};
        int index = -1;
        for (int i = 0; i < a.length; i++) {
            if (a[i] == 52) { index = i; break; }
        }
        if (index >= 0) System.out.println("Found at index " + index);
        else System.out.println("Not found");
    }
}
```

3. **Output:** `Found at index 1`
4. **Viva-Voce:** Why initialize index to -1? What does break do? What does the output mean?

Experiment No. 7.2: Bubble sort the array {3, 1, 2}

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_02_Bubble_sort_the_array {
    public static void main(String[] args) {
        int[] a = {3, 1, 2};
        for (int pass = 0; pass < a.length - 1; pass++) {
            for (int i = 0; i < a.length - 1 - pass; i++) {
                if (a[i] > a[i + 1]) {
                    int temp = a[i]; a[i] = a[i + 1]; a[i + 1] = temp;
                }
            }
        }
        for (int value : a) System.out.println(value);
    }
}
```

3. **Output:**

```
1
2
3
```

4. **Viva-Voce:** What is an adjacent pair? Why does the inner loop shrink? What is the sorted result?

Experiment No. 7.3: Multiply array values

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_03_Multiply_array_values {
    public static void main(String[] args) {
        int[] a = {2, 3, 4};
        int product = 1;
        for (int value : a) product *= value;
        System.out.println("Product: " + product);
    }
}
```

3. **Output:** `Product: 24`
4. **Viva-Voce:** Why initialize product to 1 instead of 0? What is multiplication's identity element?

Experiment No. 7.4: Print even array values

1. **Aim:** Implement the task specified in the BCS2024545 lab assignment.
2. **Program:**

```
public class Task_04_Print_even_array_values {
    public static void main(String[] args) {
        int[] a = {1, 2, 3, 4};
        for (int value : a) {
            if (value % 2 == 0) System.out.println(value);
        }
    }
}
```

3. **Output:**

```
2
4
```

4. **Viva-Voce:** What does % calculate? How do you test whether an integer is even? Does this program print odd values?